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Answer to Q1(i)

A computer is a machine that takes input, processes data, and gives output. It has four parts: Input unit which takes data from user like keyboard, CPU which does all processing work, Memory which stores programs and data, and Output unit which shows results on monitor. CPU is the most important part.

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Answer to Q1(ii)

RAM is primary memory. It is fast and expensive. When we open a program it loads in RAM. Hard disk is secondary memory. It stores data permanently. RAM is volatile means data is lost when power goes off. HDD keeps data. Examples of RAM: DDR3, DDR4. Examples of HDD: Seagate HDD, Samsung SSD.

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Answer to Q1(iii)

OS is system software that manages all resources of computer. Functions are: 1. Process management - manages all running programs. 2. Memory management - allocates memory to programs. 3. File management - handles files and folders. 4. Device management - controls printers, keyboard etc. 5. Security - protects system from unauthorized access.

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BAR001**Answer to Q1(iv)**

Multiprogramming means multiple programs are kept in memory at same time. When one program is doing I/O, CPU does work of other program. This increases utilisation. Multitasking is when we can do many tasks at same time. Like listening music and typing document. CPU switches between tasks very fast so all tasks seem to run together.

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Answer to Q2(i)

Von Neumann architecture has CPU, Memory, Input and Output connected by system bus. CPU has Control Unit and ALU. Memory stores instructions and data. Input unit takes data from user. Output unit shows result. All communicate through data bus, address bus and control bus.

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Answer to Q2(ii)

Fetch: CPU reads instruction from memory. Program Counter gives address. Instruction goes to Instruction Register. Decode: Control Unit reads opcode and finds out what to do. Execute: ALU does the operation. Result stored in register or memory. PC increments to next instruction.

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Answer to Q3(i)

255 in binary: I divide 255 by 2 repeatedly. $255=11111111$. 255 in octal: 377. 255 in hexadecimal: FF. Check: $F=15$, $FF = 15 \times 16 + 15 = 255$. Correct.

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